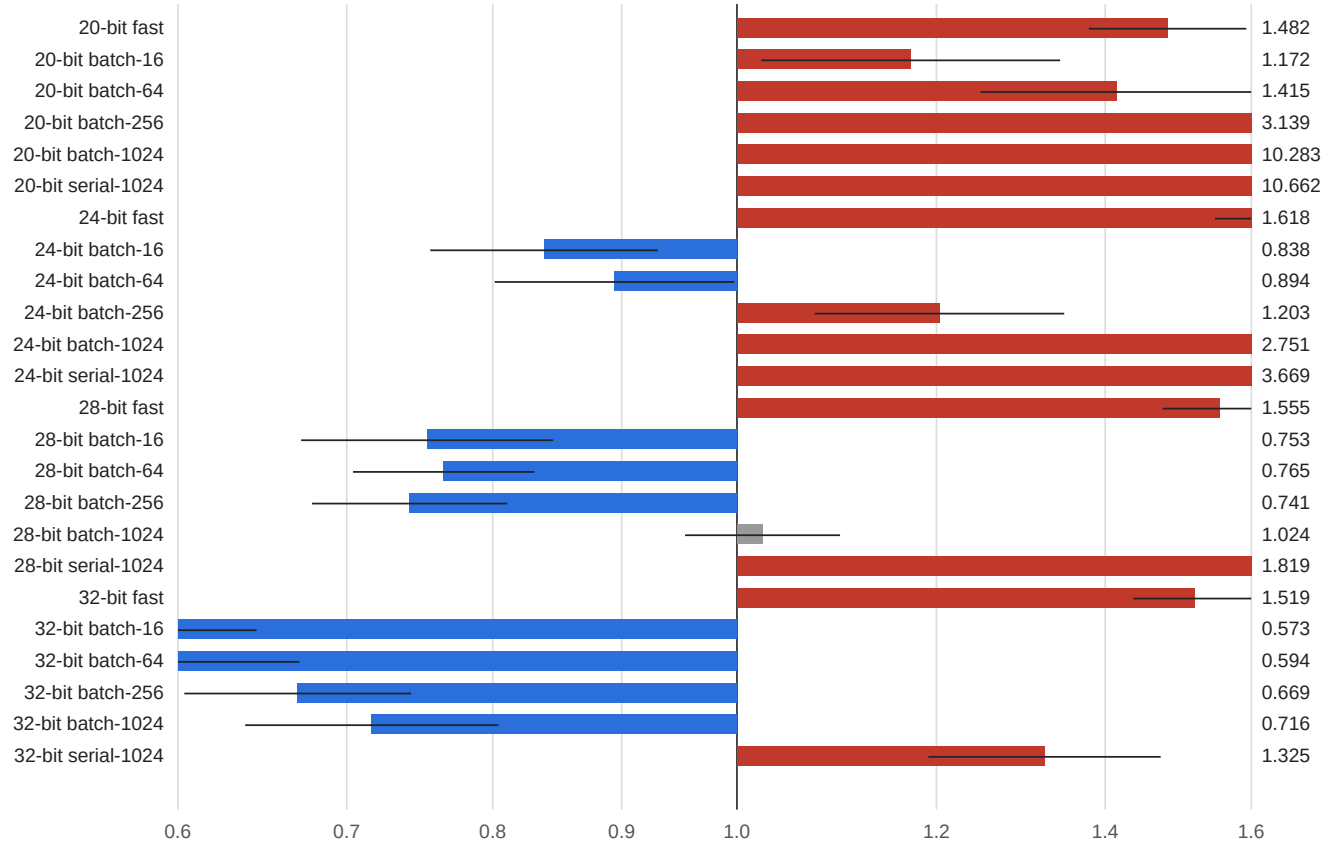
Wide target batching for prime index calculus (2026-10-28)

Lead `ICP-prime-ic-constant-engineering`: step 64, 256 and 1024 target walks in lock-step so that one field inversion serves a whole round, against the incumbent guarded walk, on the pair-table recipe (`mitm-prime-batch:m=2`). Three sessions per size in one binary, every arm alternated within each session.

Relation-phase wall time against the incumbent walk (below 1 = faster)

paired geometric mean over the same (session, target, round); bar = 95% interval; grey = interval spans 1



bits	arm	verified	mean S	S / rho.negation	median relations ms	paired S vs walk	paired relations wall vs walk	paired solve wall vs walk
20	rho-neg	36/36	1.14	1.0	nan	nan	nan	nan
20	walk	36/36	20.17	17.7	0.80	1.000	1.000	1.000
20	fast	36/36	20.17	17.7	1.15	1.000	1.482	1.141
20	batch-16	36/36	21.29	18.7	0.80	1.056	1.172	1.025
20	batch-64	36/36	23.32	20.5	1.09	1.157	1.415	1.112
20	batch-256	36/36	33.77	29.7	2.34	1.676	3.139	1.668
20	batch-1024	36/36	76.02	66.8	7.63	3.773	10.283	3.949
20	serial-1024	36/36	76.02	66.8	7.98	3.773	10.662	4.072
24	rho-neg	36/36	1.03	1.0	nan	nan	nan	nan
24	walk	36/36	21.96	21.4	5.41	1.000	1.000	1.000
24	fast	36/36	21.96	21.4	8.65	1.000	1.618	1.324
24	batch-16	36/36	22.07	21.5	4.19	1.006	0.838	0.891
24	batch-64	36/36	22.75	22.1	4.72	1.036	0.894	0.940
24	batch-256	36/36	25.77	25.1	6.34	1.174	1.203	1.107
24	batch-1024	36/36	39.51	38.5	13.57	1.801	2.751	1.932
24	serial-1024	36/36	39.51	38.5	17.80	1.801	3.669	2.404
28	rho-neg	36/36	1.03	1.0	nan	nan	nan	nan

28	walk	36/36	24.80	24.1	50.20	1.000	1.000	1.000
28	fast	36/36	24.80	24.1	77.35	1.000	1.555	1.413
28	batch-16	36/36	24.92	24.2	37.74	1.003	0.753	0.821
28	batch-64	36/36	25.54	24.8	39.16	1.030	0.765	0.825
28	batch-256	36/36	25.66	24.9	39.77	1.032	0.741	0.812
28	batch-1024	36/36	29.78	28.9	50.41	1.200	1.024	1.030
28	serial-1024	36/36	29.78	28.9	90.45	1.200	1.819	1.609
32	rho-neg	36/36	1.02	1.0	nan	nan	nan	nan
32	walk	36/36	33.01	32.4	473.33	1.000	1.000	1.000
32	fast	36/36	33.01	32.4	761.86	1.000	1.519	1.431
32	batch-16	36/36	31.08	30.5	282.24	0.940	0.573	0.643
32	batch-64	36/36	31.95	31.4	326.67	0.965	0.594	0.664
32	batch-256	36/36	33.69	33.1	341.50	1.019	0.669	0.721
32	batch-1024	36/36	34.22	33.6	379.64	1.033	0.716	0.765
32	serial-1024	36/36	34.22	33.6	693.61	1.033	1.325	1.271

Counted S includes every lane's in-flight work and the start-up of every lane, so S cannot fall with width; the measurement is the relation-phase wall time.

Findings

- **Sixteen lanes is the knee; wider does not pay.** Relation-phase wall time against the incumbent walk at 32 bits: 16 lanes 0.573, 64 lanes 0.594, 256 lanes 0.669, 1024 lanes 0.716; at 28 bits 0.753, 0.765, 0.741, 1.024. At 1024 lanes the 28-bit interval (0.954, 1.099) spans 1: no gain over the walk.
- **The shared inversion is the mechanism.** The serial control with the same 1024 walks and no shared inversion is 1.325 of the walk at 32 bits, against 0.716 with it.
- **Counted S cannot fall with width.** Every lane's start-up and the work in flight when the logarithm is found are charged: paired S against the walk at 20 bits is 1.056 (16 lanes) and 3.773 (1024 lanes).
- **Small curves lose.** At 20 bits even 16 lanes cost 1.172 of the walk's relation time: the lane start-up is not amortised over a solve of a few thousand trials.
- **Existing path to look at.** `walk-prime-fast` (native Montgomery additions, same S) is slower than the plain walk at every size: 1.48, 1.62, 1.55, 1.52 of the walk's relation time.
- **Verdict for the lead.** The predicted 2x from stepping 1024 targets in lock-step comes from a regime in which one inversion per target dominates; at 16 lanes the inversion is already shared and is a few nanoseconds a step, so the lead's wide-batching item closes as measured. The constant gain that exists is the 16-lane one (round 3), now measured at 24 and 28 bits as well.